



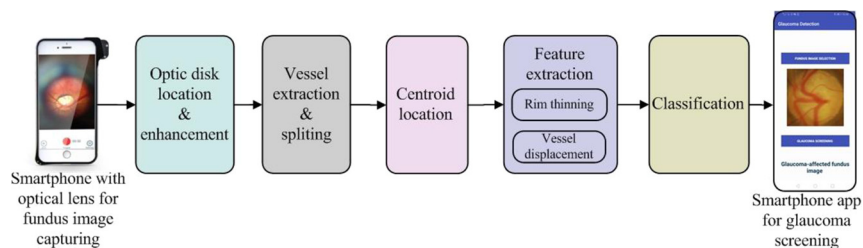
Original Article

A Fast and Accurate Method for Glaucoma Screening from Smartphone-Captured Fundus Images[☆]Y. Mrad^a, Y. Elloumi^{a,b,c,*}, M. Akil^b, M.H. Bedoui^a^a Medical Technology and Image Processing Laboratory, Faculty of medicine, University of Monastir, Tunisia^b LIGM, Univ Gustave Eiffel, CNRS, ESIEE Paris, F-77454 Marne-la-Vallée, France^c ISITCom Hammam-Sousse, University of Sousse, Tunisia

HIGHLIGHTS

- Proposition of an accurate and fast method for glaucoma screening based on the vessel displacement.
- Achievement of higher performance detection with smartphone captured fundus images.
- Implementation of the method into a smartphone to provide a mobile aided screening system for glaucoma.

GRAPHICAL ABSTRACT



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ABSTRACT

The glaucoma is an eye disease that causes blindness when it progresses in an advanced stage. Early glaucoma diagnosis is essential to prevent the vision loss. However, early detection is not covered due to the lack of ophthalmologists and the limited accessibility to retinal image capture devices.

In this paper, we present an automated method for glaucoma screening dedicated for Smartphone Captured Fundus Images (SCFIs). The implementation of the method into a smartphone associated to an optical lens for retina capturing leads to a mobile aided screening system for glaucoma. The challenge consists in insuring higher performance detection despite the moderate quality of SCFIs, with a reduced execution time to be adequate for the clinical use. The main idea consists in deducing glaucoma based on the vessel displacement inside the Optic Disk (OD), where the vessel tree remains sufficiently modeled on SCFIs. Within this objective, our major contribution consists in proposing: (1) a robust processing for locating vessel centroids in order to adequately model the vessel distribution, and (2) a feature vector that relevantly reflect two main glaucoma biomarkers in terms of vessel displacement. Furthermore, all processing steps are carefully chosen based on lower complexity, to be suitable for fast clinical screening. A first evaluation of our method is performed using the two public DRISHTI-DB and DRIONS-DB databases, where 99% and 95% accuracy, 96.77% and 97.5% specificity and 100% and 95% sensitivity are respectively achieved. Thereafter, the method is evaluated using two fundus image databases respectively captured through a smartphone and retinograph for the same persons. We achieve 100% accuracy using both databases which assesses the robustness of our method. In addition, the detection is performed on 0.027 and 0.029 second when executed respectively on the Samsung-M51 on the Samsung-A70 smartphone devices. Our proposed smartphone app provides a cost-effective and widely accessible mobile platform for early screening of glaucoma in remote clinics or areas with limited access to fundus cameras and ophthalmologists.

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* Corresponding author.

E-mail address: yaroub.elloumi@esiee.fr (Y. Elloumi).

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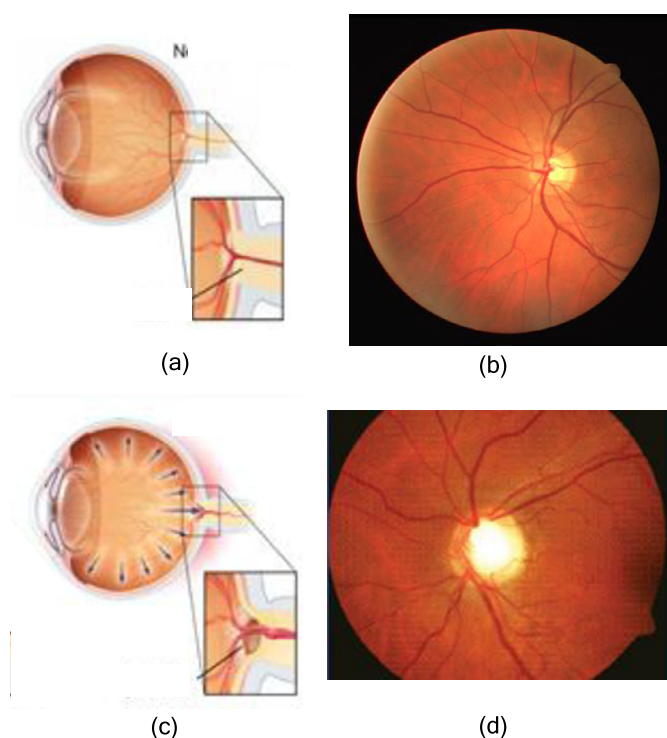


Fig. 1. (a) Optic nerve head excavation of healthy retina; (b) Fundus image of healthy retina; (c) Optic nerve head excavation of glaucomatous retina; (d) Fundus image of glaucomatous retina.

1. Introduction

Glaucoma damages the Optic Nerve Head (ONH) due to elevated or inappropriate intraocular pressure in the eye. It is estimated that patients suffering from glaucoma had reached 76 million by the end of 2020 and will reach 111.8 million by 2040 [1]. Glaucoma gradually and asymptotically causes the loss of the visual field [2]. It is often referred as a “silent thief of sight” which is noticed only when the disease progresses to a significant loss of peripheral vision.

There are two main types of glaucoma, which are the closed-angle glaucoma and the open-angle glaucoma. Rarely found, the first type is caused by a sudden blockage of eye drainage system which results in acute pain, red eye, and immediate loss of vision. The second type can be defined as an inefficiency of drainage system of the eye with time, which leads to gradual increase of fluid pressure within an eye [3]. It is considered as common, curable and painless type of glaucoma, which is addressed in this work. The main characteristics of glaucoma are the deterioration of optic nerve fibers and the degeneration of astrocytes due to a high Intra-Ocular Pressure (IOP) as illustrated in Fig. 1(c). The first characteristic causes the decrease of the thickness of Retinal Nerve Fiber Layer (RNFL). The second one leads to alter the ONH where the cup is expanded as shown Fig. 1(d) [4], and thereby decreasing the functional capability of retina.

Retinal imaging is actually considered as mainstay of the clinical care of patients with retinal diseases. Multiple techniques are used to detect retinal disorders, where the common ones are the Optical Coherence Tomography (OCT) which permit to observe the retina layers and fundus photography that allows observing the retina texture. The fundus imaging is widely used for detecting diabetic retinopathy [5], hypertensive retinopathy [6] and age-related macular degeneration. For the glaucoma disease, the fundus images are widely used for screening where the ONH and the RNFL damages can be noticed for the retina front.

The early stage of glaucoma is absolutely painless and usually manifests with no symptoms. However, the occurred changes in retinal structures lead to a gradually vision loss and eventually cause blindness. Thus, an accurate and early glaucoma detection is very crucial to stop the disease progression and to prevent the vision loss. Therefore, several studies have proposed automatic methods for glaucoma screening based on color fundus images, which differ with respect to the biomarkers addressed for glaucoma detection.

Several methods aimed to screening glaucoma based on CDR values [7,8]. In [9], the Cup-to-disc ratio (CDR) is deduced for Glaucoma detection from fundus images using the support vector machine (SVM), where an accuracy of 100% is achieved. Qurechi et al. [10] were propounded to segment the OD through a contrast enhancement and thresholding. Thereafter, the OC is segmented following a feature extraction step. Hence, the glaucoma is assessed with 96.1% accuracy and 97% specificity. In [11], an extended DeepLabv3+ architecture is employed for the CDR assessment where accuracies between 86.84% and 99.53% are reached for several public datasets. The method proposed in [12] is based on a texture-based and model-based approach to segment the OD and the OC. The glaucoma is deduced if the CDR is limited with parameters computed in terms of mean and standard deviation of CDR in healthy and glaucomatous images. A final glaucoma screening is achieved with 98% of accuracy. The method proposed in [13] aims to screen glaucoma based on RNFL degradation, for such a need, texture features are modeled through a co-occurrence matrix which is provided to backpropagation neural network, where an overall accuracy of 94.52% is achieved. Further, other methods addressed multiple glaucoma biomarkers to guaranty a reliable screening [14] [15]. Issac et al. [16] aimed to study the blood vessel distribution into the OD, the ISNT rules and CDR to screen glaucoma. The proposed method is based on an Artificial Neural Network (ANN) Classifier which achieved an accuracy of 87.5%. The method propounded in [17] proposed a new deep learning model where the output feature map is browsed to deduce CDR, RNFL and ISNT rule. The evaluation attested an accuracy of 91.51%. In [18], an extended U-Net architecture is used to segment the OD and OC in order to deduce the CDR and ISNT biomarkers, where accuracy of 0.87 with a sensitivity of 0.85 were attained.

However, the capturing process required the presence of an ophthalmologist where a worldwide deficiency is registered. Moreover, retinal imaging requires expensive conventional retinography devices who are unavailable either in rural areas of developed countries or in countries under-development. In addition, the capturing process is always insured in medical institutions which implies patient mobility. In contrast, this latter is limited due to the decreased visual acuity caused by the glaucoma disease and the older age of the most glaucoma affected people. Consequently, the proposed methods have not insured detecting glaucoma in early stage. Even have been proposed since the last twenty years, the visual acuity and blindness caused by glaucoma remain in permanent rise, which impact the patient healthcare. Moreover, a large worldwide population that reached the advanced stage, required a high cost therapy which achieve 2,511 \$ and 803 € respectively in United States and Europe [19].

Actually, several mobile devices for fundus image capturing have been suggested such as D-eye and iExaminer, which allow using smartphones in eye disease screening, taking advantage from their mobility, low-cost and high capabilities in terms of processing and storage [20–22]. It has been deduced that Smartphone Captured Fundus Images (SCFIs) have an acceptable quality compared to the images captured by ophthalmoscopes, where both fundus images, in Fig. 2, are captured from the same eye using respectively a retinograph and an optical lens-associated smart-

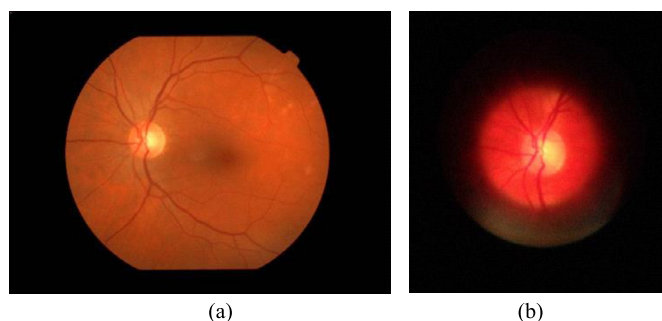


Fig. 2. Fundus images: (a) captured with retinograph; (b) captured with optical lens-associated smartphone.

phone. In [23], the D-eye showed considerable accuracy of CDR measurements compared to direct ophthalmoscopy. In addition, the D-eye gadget offered high performance for manually detecting fundus abnormalities compared to indirect ophthalmoscopy, with specificity of 0.97 and 0.98 for detecting the eye abnormalities and measuring the optic nerve size. Nevertheless, the handheld capturing leads to a blurred and noised illustration of retinal components. Hence, the SCFIs have a moderate quality with respect to the ones captured with conventional retinography, where the retina is illustrated with a sharpness quality and a raised resolution. Moreover, the SCFIs are captured with a Reduced FOV in comparison to the classical fundus images, as shown in fundus images of Fig. 2. The works in [24] proved that detection performances of ocular diseases decrease drastically in terms of both blur and limited FOV.

Some work has been interested into proposing a mobile MAS system for glaucoma to achieve higher performance detection with lower processing complexity. The work proposed in [12] put forwarded a computationally-efficient algorithm for further implementation on mobile devices. However, the whole implementation requires 11 seconds when executed in a desktop architecture. The work proposed by [25] was indicated that the tele-glaucoma technology demonstrated moderate agreement on its ability to diagnose glaucoma and achieved 79% of specificity and 83% of sensitivity. The work described in [26] suggested a mobile application for glaucoma screening which is ensured with accuracy of 0.7625 in 9.16 seconds. Martins et al. [18] have propounded a method for diagnosing glaucoma with an accuracy of 0.87 and a sensitivity of 0.85 which is executed into a smartphone under two seconds. Aside the elevated computational performance, we distinguished that none of those works had addressed the problems of limited quality of SCFIs.

Therefore, our main objective is to propose a novel method for glaucoma screening from smartphone-captured fundus images. The implementation of the method on a smartphone associated with a mobile device for fundus image capturing leads to provide a Mobile Aided Screening (MAS) system for glaucoma. Our first challenge consists in ensuring a higher accuracy even with moderate quality of SCFIs. The second challenge is to perform the detection into a lower computational complexity in-order to implement the method into a smartphone. Our main approach consists in addressing the retinal component and the glaucoma biomarkers where the extraction remains robust whatever the SCFI quality is. In addition, all processing steps must have lower complexities even for retinal component location or for biomarker deduction. For such need, our method idea corresponds to studying the vessel distribution inside the OD and retrieves NRR and the ISNT rules from vessel distributions to deduce the glaucoma disease. This work is a part of a mobile computer-screening system for ocular pathologies which we already developed [27–29].

The rest of the paper is expended to present the suggested method. The second section describes the automated method for glaucoma detection. The experimental evaluation of the detection performance and the execution time is presented in section 3. The conclusion and some future work are detailed in section 4.

2. Fast and accurate method for glaucoma screening

2.1. Approach principle

Our challenges consist in insuring an accurate and robust method even with a moderate image quality. Furthermore, all processing steps must be carefully chosen based on lower complexity, to be suitable for mobile implementation. The lesions caused by glaucoma disease are focused on the OD region which assists the glaucoma diagnosis, in contrast to the other ocular pathologies such as diabetic retinopathy [5] and hypertensive retinopathy [30,31]. Therefore, several existing methods have succeeded to achieve higher glaucoma detection performance either through DL-based methods, classical machine learning or image processing based ones [10]. However, the complex computation of DL architectures leads to higher execution times. Those computational performances will be downgraded when executed in smartphone devices that generally have lower processing capacities. A second constraint for using DL architecture consists in using a large database to perform a reliable training. In our case, we distinguished a varied deterioration on SCFI quality, which involves a wider dataset. In contrast, we are constrained with a lower size of SCFIs. To reach the targeted challenges, we proceed to assess the glaucoma through a machine learning-based method. Our main idea is to address the retinal component and the glaucoma biomarkers which insure higher detection whatever the SCFI quality is.

At first, glaucoma damages the optic nerve where the main morphological changes occur inside the ONH. Indeed, glaucoma screening from the ONH brings to higher performance detection and low complexity processing [32–35] in contrast to the methods exploring the whole retina, as synthesized in the comparative study performed in [36]. Moreover, SCFIs are characterized by a reduced field of view against the classical fundus images, hence avoiding presenting the papillary atrophy located outside the ONH. Therefore, the first step of our method consists in locating the OD, as detailed in section 2.2.

Thereafter, we study the impact of glaucoma on the retinal components in order to identify the adequate ones for screening, taking into account the moderate quality of SCFIs. A healthy optic nerve corresponds to a tiny excavation. However, glaucoma damages the optic nerve fibers, thus expanding the excavation, as it can be deduced when comparing Fig. 1(a) and Fig. 1(c). Similarly, the OC appears with a reduced size in a healthy fundus image. For a glaucoma-affected one, the excavation growth leads to increase the OC size, while the OD size remains stationary, as respectively distinguished from the fundus images in Fig. 1(b) and Fig. 1(d). Elsewhere, the excavation growth does not occur similarly in all directions where the glaucoma leads to destroy the ISNT rule. With respect to this indication, several methods have segmented the OD and the OC to deduce whether the CDR or shape of the NRR to detect glaucoma.

In a healthy retina, vessels spread from the middle of the excavation, as shown in Fig. 1(a). The vessels propagate beyond the large NRR, which appear with an equitable distribution in the OD, as modeled in Fig. 1(b). Otherwise, the rise in the optic nerve fiber damage leads to join the vessel to the internal border of the excavation, as we can see in Fig. 1(c). The perpendicular visual field

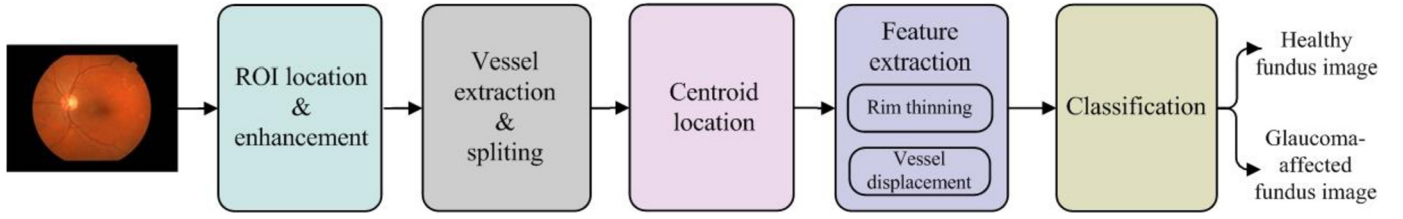


Fig. 3. Flowchart of proposed method for glaucoma screening.

to the retina leads to model vessels as shifted to the OD border, as depicted in Fig. 1(d). Accordingly, the optic nerve fiber damage similarly affects the OC size and the vessel distribution, where both are able to reflect glaucoma severity. In the same vein, several methods have identified the glaucoma through segmenting and locating retinal vessels.

SCFIs are always blurred and noised due to the handheld aspect, which have a moderate quality with respect to classical images. Degradation principally affects the contrast of retinal components with respect to their neighborhood. However, the OD and the OC have close intensity and blurred borders, which leads to alter at once the OD and OC segmentation. By applying their ratio, such drop in performance considerably fakes glaucoma screening. In contrast, the vessel tree remains sufficiently represented in the OD. The vessel distribution is obviously distinguishable, despite the moderate quality.

For this purpose, the main idea of our method consists in screening glaucoma from SCFIs based on blood vessels. To guarantee the method robustness, our approach will be based on evaluating the distribution of the whole vessel tree rather than based on each vessel separately. Therefore, the vessel tree is segmented and partitioned into quadrants with respect to the ISNT notion, as described respectively in sections 2.3 and 2.4. Then, a position marker of vessel distribution on each quadrant is located, which called centroid. After that, our second main contribution leads to extract a vector that relevantly reflect two main glaucoma biomarkers in terms of vessel displacement. The feature vector is then provided to a classifier to deduce glaucoma, as summarized in the method flowchart in Fig. 3.

2.2. OD extraction and enhancement

The method proposed in [37] ensured locating the ONH based on circularity and intensity through applying the radon transform. The method achieved higher accurate performance among the existing methods [38,39], where 100% accuracy was performed to locate the ONH in the public DRIVE database. We note that this method can be configured to locate a sub-image having the same OD size. Afterwards, this method was extended to provide a mobile computer aided system for ONH detection [27]. Processing was optimized to reduce complexity, where the same detection accuracy was performed in reduced time when executed in a smartphone device. Therefore, the method suggested in [27] would be implemented, which extracted the OD sub-images.

In addition, glaucoma screening requires detecting overlapped retinal components. However, SCFIs are characterized by a non-balanced contrast, which result in failed detection. Consequently, the second step leads to enhance the sub-image contrast with respect to the blood vessels. We proceed to extract the green channel which reveals a better contrast between the retinal structures. Next, the CLAHE algorithm is applied to ensure a better contrast equalization into the ONH region, where the results of each step are illustrated in Fig. 4.

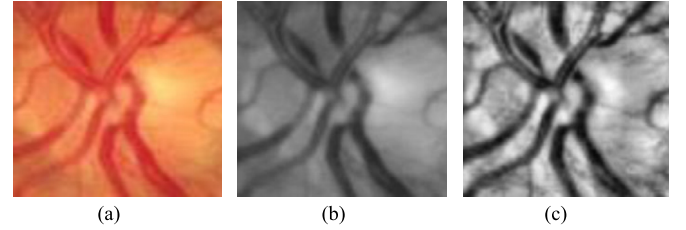


Fig. 4. (a) Healthy OD; (b) Green channel, (c) OD after applying CLAHE algorithm.

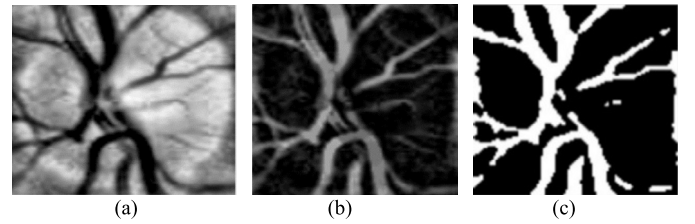


Fig. 5. Blood vessel segmentation: (a) green channel, (b) black-hat inversion, (c) Otsu segmentation + MedianBlur.

2.3. Vessel segmentation and splitting

We proceed to segment the vessel tree into the located OD. The first step consists in inverting the OD brightness in order to model the blood vessel with higher intensity. Within this objective, the “black-hat” morphological operator is applied on the green channel f . The processing consists in applying opening to the input image f using the structuring element b and then subtracting f , as indicated in equation (1). The input and output are respectively depicted in Fig. 5(a) and Fig. 5(b).

$$T_b(f) = f \bullet b - f \quad (1)$$

Then, the Otsu technique is applied to distinguish vessel from the OD background [40]. This method searches for the threshold that minimizes the intra-class variance between the vessel and background class, defined as a weighted sum of variances $\sigma^2(t)$ of those classes, as cited in equation (2).

$$\sigma^2(t) = w_{bg}(t) \times \sigma_{bg}^2(t) + w_{vessel}(t) \times \sigma_{vessel}^2(t) \quad (2)$$

where $w_{bg}(t)$ and $w_{vessel}(t)$ are the Weights and $\sigma_{bg}^2(t)$, $\sigma_{vessel}^2(t)$ are the probabilities of both classes separated by a threshold t .

Finally, the “Medianblur” filter is used to reduce noise, which can be caused by small blood lesions [41], as shown in Fig. 5(c). The processing principle corresponds to replace the pixels $f(x, y)$ by the median value of their neighborhoods. The pixels belonging to a $w \times w$ mask are ranked in order of their gray scale where the median value is selected to replace the initial pixel value, as mentioned in equation (3).

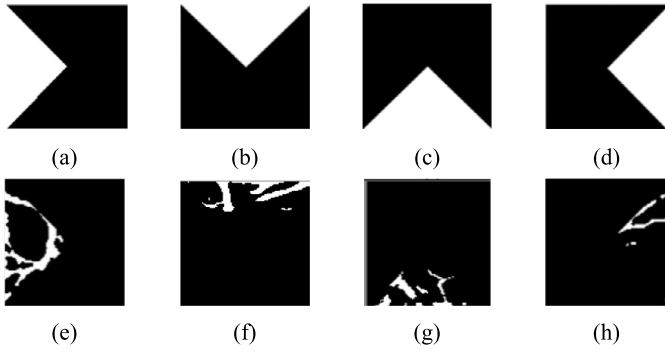


Fig. 6. (a-d): Masks used for quadrant partitioning; (e-h): blood vessels of each quadrant.

$$g(x, y) = \text{median}\{f(x, y), (i, j) \in W\} \quad (3)$$

The vessel tree displacement caused by glaucoma is developed mainly in superior, inferior and nasal quadrants, where the temporal quadrant contains less vessels than the other ones [42,43]. Thereby, vessels should be partitioned on their quadrants to deduce the glaucoma displacement. Within this objective, we proceed to split the segmented vessels into four quadrants through superimposing four different masks, illustrated in Fig. 6(a)-Fig. 6(d), to the OD where the results are respectively depicted in Fig. 6(e)-Fig. 6(h).

2.4. Centroid location

It is essential to define a parameter that can reflect vessel displacement into quadrants. For each quadrant, the structure of blood vessels is presented as the white pixel in the region. Our main idea consists in computing a centroid reference point for vessels belonging to the same quadrant. This point is determined based on the coordinates of vessel pixels or the white pixels. Then, glaucoma is deduced based on the centroid remoteness.

In order to reduce processing complexity, we propose a novel approach to locate the centroids as indicated in Algorithm 1. First, for the vessel pixels of each quadrant, the sum of their horizontal coordinate Q_x and vertical coordinate Q_y are computed, as indicated respectively in equations (5) and (6). The centroid horizontal coordinate C_x (resp. vertical coordinate C_y) corresponds to the ratio between Q_x (resp. Q_y) and the number of vessel pixels N_{VP} , as indicated in equations (7) (resp. in equation (8)).

Algorithm 1 Centroid location.

INPUT: image of segmented vessels I_{vessel} , nbr and nbc respectively the row and column number of I_{vessel}
OUT: the centroid coordinates
Begin
 $N_{VP} \leftarrow 0$
 $Q_x \leftarrow 0$
 $Q_y \leftarrow 0$
 $C_x \leftarrow 0$
 $C_y \leftarrow 0$
For $x=0$ **to** $\text{nbr}-1$ **do**
 For $y=0$ **to** $\text{nbc}-1$ **do**
 If $I_{\text{vessel}}(x_v, y_v)$ **then**
 $N_{VP} \leftarrow N_{VP} + 1$
 $Q_x \leftarrow Q_x + x_v$
 $Q_y \leftarrow Q_y + y_v$
 End if
 End for
End for
 $C_x \leftarrow (Q_x / N_{VP})$
 $C_y \leftarrow (Q_y / N_{VP})$
End

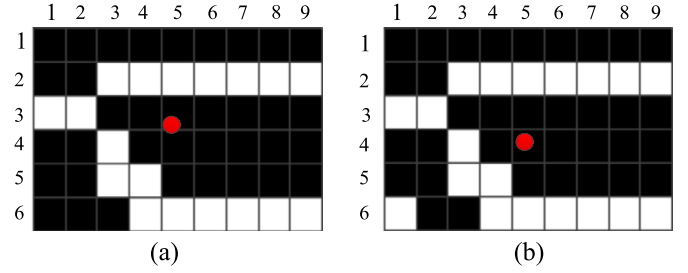


Fig. 7. Illustration of the centroid position. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

$$N_{VP} = \sum_{i=0}^{\text{nbc}} \sum_{j=0}^{\text{nbr}} I_{\text{vessel}}(x_v, y_v) \quad (4)$$

$$Q_x = \sum_{i=1}^{N_{VP}} x_v \quad (5)$$

$$Q_y = \sum_{i=1}^{N_{VP}} y_v \quad (6)$$

$$C_x = \frac{Q_x}{N_{VP}} \quad (7)$$

$$C_y = \frac{Q_y}{N_{VP}} \quad (8)$$

where $I_{\text{vessel}}(x_v, y_v) = (255, 255, 255)$.

For example, Fig. 7(a) and Fig. 7(b) illustrate the centroid locations in respectively healthy and glaucoma ODs, respectively. A vessel bifurcation is illustrated in Fig. 7(a) where the centroids are modeled by red circles having $C_x = 3.88$ and $C_y = 5.22$. To approve the robustness of the centroid location, we added a noise pixel $I_{\text{vessel}}(1, 6)$, where we deduce that centroid location was slightly modified whose coordinates become $C_x = 3.73$ and $C_y = 5.26$.

2.5. Feature extraction

2.5.1. Neuro-retinal rim thinning

In a healthy fundus image, the rim has a large board through the OD. The glaucoma involves increasing the OC and hence reducing the NRR. This criterion remains very helpful for clinical staff to diagnose glaucoma [38]. Within this context, the method described in [44] computed the rim-to-disc ratio in order to deduce glaucoma. In [45], the rim area of each quadrant is figured out in order to compute a ratio that reflects glaucoma.

In fact, a significant correlation is distinguished between the blood vessel distribution into quadrants and the NRR lost [45]. In a healthy fundus image, the reduced size of excavation leads to merging blood vessels to the OD center. Consequently, centroids appear as clustered in the OD middle region. The excavation growth caused by glaucoma shifts blood vessels away from the OD center until getting closer to the OD borders [42,16], such as the difference between fundus images in Fig. 1(b) and Fig. 1(d). With respect to the excavation growth, the centroid positions become spaced.

Subsequently, we aim to provide novel features that reflect the centroid remoteness. Since the four zones are affected on both extremities of the perpendicular axis, we quantify the centroid remoteness through Euclidian distances between each neighbor centroid. The temporal centroid escapes to a similar shifting since it contains less vessels. Therefore, we choose to overpass the temporal quadrant in order to guarantee higher accuracy detection. Accordingly, we consider two features which correspond to the

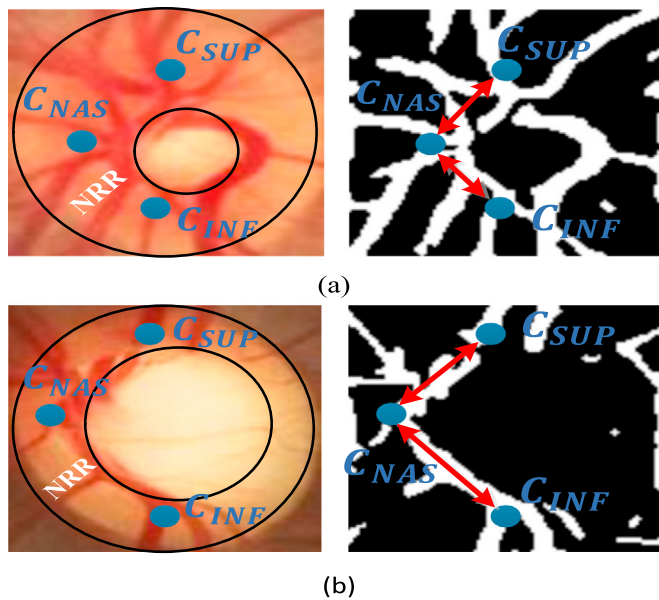


Fig. 8. Centroids of blood vessels located in different zones of OD: (a) healthy OD, (b) glaucoma affected OD.

distances between superior-nasal and inferior-nasal centroids, as illustrated with red arrows in Fig. 8, where the distances in the healthy OD in Fig. 8(a) appear clearly smaller than the ones in the glaucoma-affected OD in Fig. 8(b).

Fundus images are illustrated in Fig. 8 where the centroids are always focused inside the NRR either in the healthy or the pathological case. For the healthy OD shown in Fig. 8(a), a thick NRR correlate with lower distances $C_{NAS} - C_{SUP}$ and $C_{INS} - C_{SUP}$ equals to 30.11 and 25.02. In contrast, the thin NRR of the glaucoma-affected retina shown in Fig. 8(a) leads to raised distances $C_{NAS} - C_{SUP}$ and $C_{INS} - C_{SUP}$ equal to 52.26 and 57.68. Those significant differences on values allow guarantying an accurate glaucoma screening.

2.5.2. Vessel displacement in relation to the ISNT rule

In a healthy OD, the inferior quadrant contains the thickest rim, followed by the superior, nasal and temporal quadrants, as known as the ISNT rule [46]. Similarly, inferior and superior quadrants include the wide portion of the vessel tree [43]. For a glaucoma-affected fundus image, the optic nerve excavation growth leads to expand the OC. The rim thinning does not occur similarly in all directions where rim thicknesses into quadrants changes differently. The glaucoma involves violating the ISNT rule where the excavation expands principally in the inferior and superior quadrants. As a result, the thinning leads to shift the blood vessels from inferior and superior regions towards the nasal region [43]. Consequently, the centroids of inferior and superior quadrants remain closed to the OD border. In contrast, the added vessels to the nasal quadrant arise from inferior, superior and temporal neighborhood quadrants. The shifted vessels significantly affect the distribution of vessel density, and so the nasal centroid location. Hence, even with a thinned rim, the nasal centroid is not always relocated to be near of OD border. Similarly, the reduced vessel density in a temporal quadrant and the vessel displacement in the direction of the OD center do not allow modeling the temporal centroid displacement in terms of glaucoma.

Accordingly, we extract two features that reflect the ISNT, which correspond to two Euclidian distances between inferior and superior centroids and their corresponding borders, as indicated with red arrows in both images of Fig. 9. For the healthy OD illustrated in Fig. 9(a), the distance between C_{SUP} , C_{INF} and their correspond-

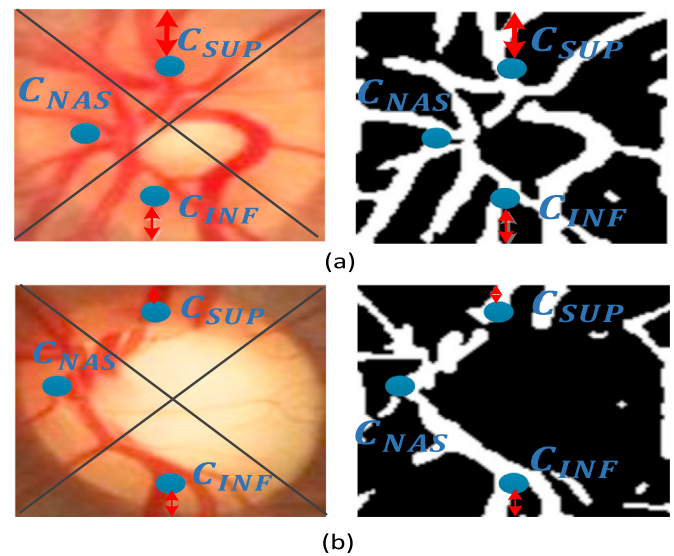


Fig. 9. (a) Healthy OD, (b) glaucoma affected OD.

ing borders are respectively equal to 16.22, 13.10, where the same distances are reduced to 10.04, 11.14 in the case of the glaucoma-affected OD shown in Fig. 9(b).

2.6. Classifier

The Support Vector Machine (SVM) classifier is a supervised learning method, used in classification and regression problems. It attempts to fit a hyperplane through the feature space in order to discriminate between classes provided as groundtruth data on the training process. The SVM classifier is frequently employed for the pathology detection method, especially for binary classification, which achieves higher performance detection. This higher accuracy is ensured even with a reduced training dataset. Moreover, the SVM classifier is characterized by lower complexity whatever the dataset size is. For this purpose, we choose the SVM as a classifier on our proposed method. The SVM consists in applying a non-linear transformation $\phi(x)$ to allocate each input vector $x = (x_1, x_2, x_3, x_4)$ on its corresponding hyperplane $h(x)$.

The kernel function is used to map the feature points to a higher dimension [47]. Since features are not linearly separable, the RBF kernel is employed, which is chosen based on the accuracy and the execution time. Elsewhere, the SVM is used with a gamma value equal to 0.1 and a C value equal to 1. The values are selected using the grid research method [48], which consists in an experiment study where the parameter choice is conducted based on detection performance.

2.7. Computational complexity

In this section, the complexity of the proposed method for glaucoma screening is studied. Firstly, the CLAHE algorithm, the “Medianblur” filter and the Otsu technique require $(n \times n)$ times for each processing, where $(n \times n)$ is the OD sub-image size. Thereafter, the nested loops of the centroid location algorithm consist in browsing all pixels of the segmented vessels image which requires $(n \times n)$ iterations to be done. The algorithm is called to locate the nasal, inferior and superior centroids, hence requiring the same computational complexity. Then, the NRR features correspond to the Euclidian distances between superior-nasal and inferior-nasal centroids which relies on $O(1)$ constant complexity. Similarly, the ISNT features correspond to Euclidian distances between centroids and their corresponding borders, which requires a $O(1)$ constant

Table 1
Databases used in proposed method.

Database	Number of fundus images	
	Healthy	Glaucoma images
DRIONS-DB	40	20
DRISHTI-GS	31	70

complexity. Finally, the classification ensured by the SVM classifier is equal to $\max((f \times d), d^2)$ [49], where f is the number of features and d is the image number used for training, hence requiring constant computational complexity. Consequently, the proposed method for glaucoma screening relies on low complexity of $O(n \times n)$.

3. Experimental results

3.1. Evaluation of glaucoma screening method

3.1.1. Dataset and evaluation metrics

To test and evaluate the proposed method, two publically available datasets are selected which are frequently used in several methods for glaucoma assessment. The first one is the DRISHTI-DB dataset which contains a total of 101 images provided by the Medical Image Processing (MIP) group, IIIT Hyderabad. All images were taken with dilated eyes, centered on the OD with a field of view of 30 degrees and a dimension of 2896×1944 pixels. The second dataset is the DRIONS-DB [50,51] which contains 110 digital retinal images including 50 normal images and 60 glaucoma images with a dimension of 600×400 pixels. The fundus images were acquired from patients approximately 53 years of age, approximately centered on the ONH [38]. We note that DRIONS-DB contains suspect fundus images which are avoided in our experimentation. The overall healthy and glaucoma-affected images are mentioned in Table 1.

The performance of our method is evaluated through the accuracy, sensitivity and specificity metrics [52,12], which are computed as indicated in equations (9), (10) and (11).

$$\text{Sensitivity} = \frac{TP}{TP + FN} \quad (9)$$

$$\text{Specificity} = \frac{TN}{TN + FP} \quad (10)$$

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + TN + FN} \quad (11)$$

Where TP (True Positive) is the number of images classified correctly as glaucoma-affected, TN (True Negative) is the number of images classified correctly as healthy, FP (False Positive) is the number of healthy images identified as glaucomatous and FN (False Negative) is the number of glaucoma-affected images classified as healthy. To correctly evaluate our contribution, those metrics are computed after locating the OD using the method detailed in [27].

3.1.2. Performance evaluation of glaucoma screening method

This method is evaluated using each dataset separately, where all fundus images are used for training and testing steps. The experimentation shows that our method achieves 99% accuracy, 100% sensitivity and 96.77% specificity when the evaluation is performed using the DRISHTI-DB database. Furthermore, when the evaluation is performed using DRIONS-DB database, our proposed method provides 95% of accuracy, 95% of sensitivity and 97.52% of specificity, as shown in Table 2. Accordingly, the higher detection performance on both datasets confirms the accuracy of our suggested method for glaucoma screening.

Table 2
Performance of the proposed method.

DB	Accuracy	Specificity	Sensitivity
DRISHTI-DB	99%	96.77%	100%
DRIONS-DB	95%	97.5%	95%

Table 3
Performance comparison with state-of-the-art methods.

Method	Accuracy	Sensitivity	Specificity
[53]	82.20%	–	–
[54]	–	95.52%	–
[55]	76.77%	–	–
[56]	95%	100%	90%
[12]	98%	100%	94.4%
Proposed method	99%	100%	96.77%

As the DRISHTI-DB is widely used for evaluating several automated methods of glaucoma screening, we compare our method with recent ones where accuracy, sensitivity and specificity are depicted in Table 3. It is obviously distinguished that our method outperforms several existing state-of-the-art glaucoma detection methods, and is tabled among the most highly performant ones.

3.2. Performance evaluation with respect to SCFIs

3.2.1. Dataset

The second experimentation consists in evaluating the suggested method using SCFIs. Within this objective, we evaluate our method using the RIAMP dataset proposed in [57], which contains healthy and glaucoma-affected fundus images acquired from 16 people. For each person, a first fundus image was captured with Topcon retinal camera and three others with the optical lens “iExaminer” snapped to a smartphone iPhone 4S. For instance, fundus images captured from healthy and glaucomatous retinas are shown respectively in first and second lines of Fig. 10. This dataset allows demonstrating that images acquired by mobile phones are applicable for the automatic detection of the OD and the OC.

The classical fundus images are provided to an ophthalmologist to generate the ground truth. Some retinas are considered as suspect, whether for classical or smartphone captured fundus images, which requires additional diagnosis to deduce if the patient is glaucoma-affected. Therefore, suspect images are avoided to keep only healthy and glaucoma-affected ones, which correspond respectively to two and nine people, for both classical and SCFI databases.

We notice that three smartphone captured fundus images are acquired for each person. Therefore, to have a stability between classes and achieve a sufficient dataset size for training, two images for each glaucoma-affected person are identified and added, based on their better quality. The structure of both datasets are detailed in Table 4.

3.2.2. Performance evaluation

To evaluate the robustness of our method, the experimentation consists in evaluating our method using each dataset separately and then comparing the achieved detection performance. For each dataset, the training and testing processes are performed using the same image set. Thereafter, the metrics of accuracy, sensitivity and specificity are computed to evaluate the method for each dataset, which are depicted in Table 5.

Based on the experimental results, we distinguish that our proposed method correctly detects all fundus images. Furthermore, the same detection performance is ensured even with smartphone captured fundus images. Accordingly, this experiment proves the robustness of the suggested method with respect to the moderate

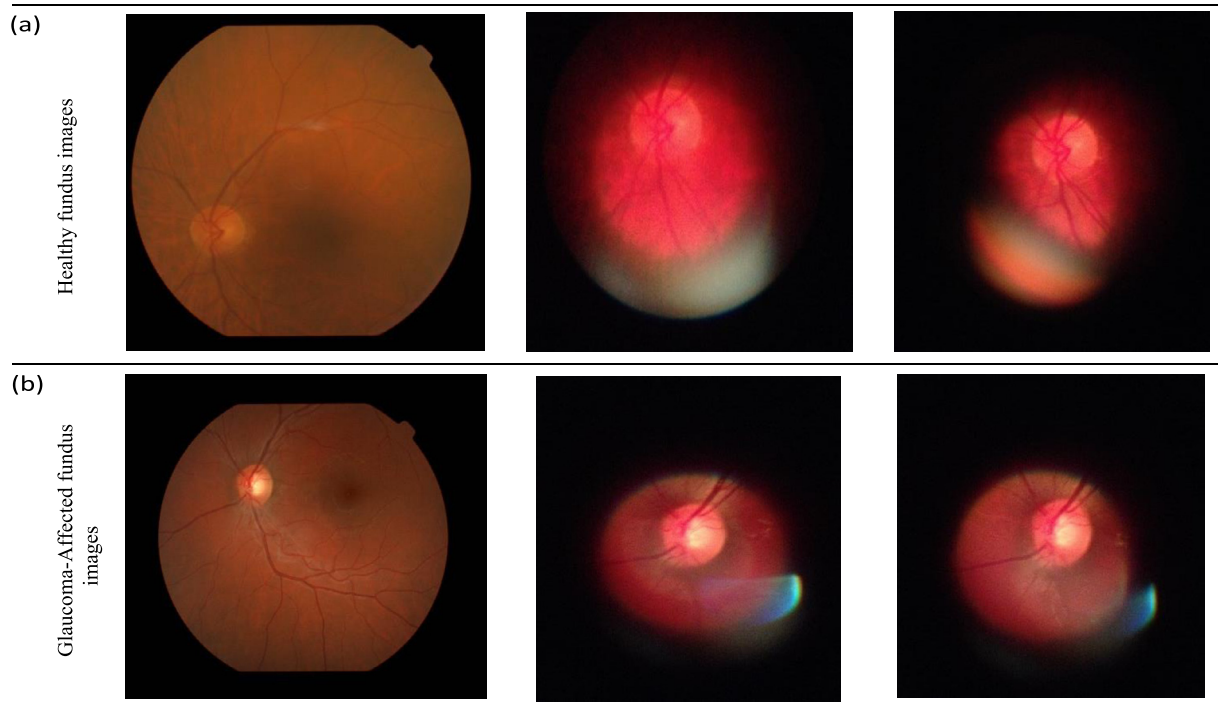


Fig. 10. Fundus images: (a) captured with Topcon retinal camera; (b) captured with iExaminer.

Table 4
Selected images from RIAMP database.

	Healthy images	Glaucoma images
Standard images	9	2
Mobile images	9	4

Table 5
Detection performances of our method in terms of datasets.

	Accuracy	Sensitivity	Specificity
Standard images	100%	100%	100%
Mobile images	100%	100%	100%

quality caused by the mobile device capturing. Hence, we can deduce that our method is adequate to be implemented it on a MAS system for glaucoma.

3.3. MAS system for glaucoma as smartphone app

The whole method is implemented as an app on an android operating system through the android SDK. This latter is a software development kit which provides several functions to fully use the system architecture and components. All image processing and machine learning steps are implemented using the “Open Source Computer Vision (OpenCV)” library [58,59]. This library has more than 2,500 optimized algorithms, which includes a comprehensive set of both classic and state-of-the-art computer vision and machine learning algorithms. It is developed using C++ and can be supported by several operating systems and interfaced through many languages such as Python, JAVA and MATLAB. The OpenCV functions are implemented on an Android app using the Android Native Development Kit (NDK) [60], which allows compiling native application codes such as C/C++ for Android applications to run on the Dalvik virtual machine. With this method, native codes can be used repeatedly and the execution speed can be faster. The Android NDK provides the native API compiler system and packages the native codes into APKs by integrating JAVA Native Interface (JNI) [61]. A graphical use interface is developed where the

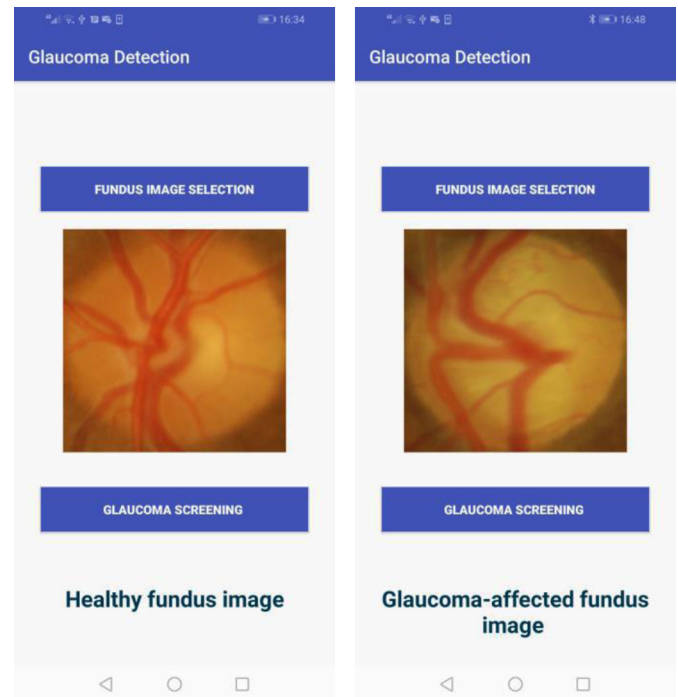


Fig. 11. Graphical Use Interfaces of the application.

first step consists at uploading a fundus images from the gallery through the button entitled “Fundus image selection”. Thereafter, the whole method is run through a button called “glaucoma detection”. The detection result is displayed below the button indicating the provided class, as shown in Fig. 11.

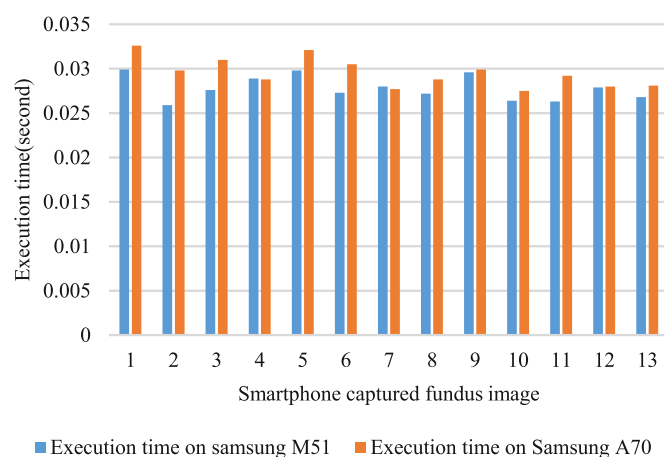
3.4. Computational performance of MAS system

In this section, we evaluate the execution time of our method when implemented in the smartphone as an app to guaranty a

Table 6

Execution times in terms of hardware and software tools of glaucoma detection methods.

Glaucoma detection methods	Execution time (in seconds)	Hardware tools	Software tools
[62]	25	core i7, 3.4 GHz RAM: 4 GB	MATLAB
[54]	19.44	Intel core i7, 3.40 GHz	MATLAB
[32]	18	Intel Core i3, 3.5 GHz RAM: 6 GB	Python 2.7 & OpenCV 2.4
[12]	11	Intel Xeon, 3.3 GHz, RAM: 8 GB	Python environment with Anaconda
[26]	9.16 ± 0.78	Cloud platform	- -
[7]	8.1	quad core, 2.66 GHz	MATLAB
[63]	2.4	Processor: NVIDIA Tesla P40 GPU	Python based on Keras with Tensorflow backend
[18]	<2	Samsung S8, Quad core, Up to 2.3 GHz, GPU Mali G71	Python & TensorFlow Lite
[33]	from 0.12 to 0.17	dual core, 3.0 GHz, RAM: 3 GB	-
Proposed method	0.027	Samsung M51, Octa core, 2.2 GHz, RAM: 6 GO	Android 10 Q & OpenCV 3.4
	0.029	Samsung A70 Octa-Core, 2 GHz RAM: 6GO	Android 9.0 & OpenCV 3.4

**Fig. 12.** Execution times in terms of smartphone captured fundus images.

reliable evaluation, 10 SCFIs are randomly selected and two smartphones with different material architectures, as mentioned in the last line of Table 6, are used. The execution times of the whole method are depicted in Fig. 12. It is deduced that the average execution times are equals to 0.027 and 0.029 seconds, respectively for the Samsung-M51 and Samsung-A70. Similarly, smaller differences of 10^{-3} and 2.15×10^{-3} are registered whether for healthy or glaucoma affected images. The achieved execution times are perfectly

adequate with the clinical context for ensuring glaucoma screening.

Thereafter, we compare our method to the state-of-the art ones where the execution times and hardware and software tools are respectively mentioned in the second, third and fourth columns of Table 6. As mentioned in the introduction, several methods do not focus on computational performance with execution times above 10 seconds in several methods. Furthermore, our smartphone-based method ensures the faster glaucoma screening, jointly with the method proposed in [33] which is performed in a desktop architecture having a processor with a 3 GHz of frequency.

3.5. Discussion and synthesis of MAS system

In this work, we proposed a glaucoma screening method from SCFIs. The main idea is based on deducing the targeted disease through the vessel displacement inside the OD quadrants. The novelty of our method consists in selecting a reduced set of features that reflect adequately the glaucoma disease, where the NRR thinning and the ISNT rule are deduced based on distances between vessel centroids. This contribution allowed to achieve the challenge that correspond to guaranty a higher accuracy even with moderate quality of SCFIs into a reduced execution times.

However, the method failed to classify some fundus images which are shown in Fig. 13. The first image corresponds to a false negative screening. The retina is in the glaucoma early stage where the excavation is slightly expanded and so vessel tree is still condensed in the OD middle. Therefore, C_{INS} and C_{SUP} centroids are located far from their corresponding borders where the distances are respectively equal to 17.1903, 16.2719. The second and third fundus images of Fig. 13 represent a false positive screening. The elliptical shape of the OD in the second image avoided to extract correctly the ROI which led to a wrong vessel splitting into quadrants and hence to a false screening. Of the third image, the lower contrast involved a blurred vessel segmentation which distorted the centroid location. To work around this issue, the OD location and vessel segmentation processing may be updated which have been addressed by several recent works.

Among methods aiming for mobile glaucoma screening, the proposed method outperforms the state-of-the-art methods [25, 26, 18] having an accuracy between 0.76 and 0.87, by achieving a higher performance screening. In addition, the screening time is considerably enhanced which reduced from 9.16 and 2 seconds in [26, 18] to 0.027 second.

4. Conclusion

In this paper, we have put forward a method for glaucoma screening dedicated to SCFIs. The method principle consists in identifying the main glaucoma proprieties that rely on in the OD, reflecting proprieties through carefully selected features and providing them to a supervised classifier. Our method achieves detection with 99% accuracy, 100% sensitivity, and 96.77% specificity when evaluated using the DRISHTI-DB. The method implementation through an app leads to a mobile computer-aided system for glaucoma detection. The experimentation confirms that our method ensures performant detection even when using SCFI. Besides, the glaucoma screening is performed on 0.027 and 0.029 second which is adequate for clinical issues.

In future work, we aim to extend our method in order to provide the severity stage of glaucoma screening. In addition, other ocular pathologies can be detected such as diabetic retinopathy and age macular degeneration. Moreover, the smartphone app can ensure uploading the fundus images to be transferred to ophthalmologists to anticipate medical care and therapy.

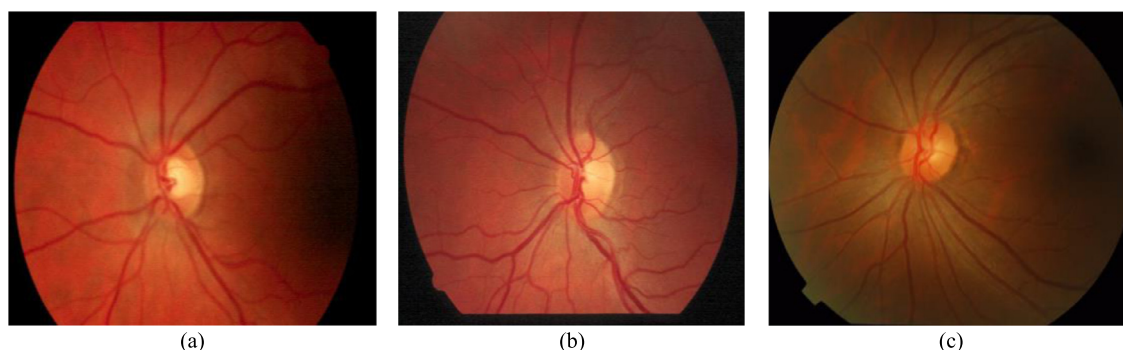


Fig. 13. The incorrectly classified Fundus images.

Informed consent and patient details

The authors declare that this report does not contain any personal information that could lead to the identification of the patient(s).

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CRediT authorship contribution statement

Y. Mrad: Conceptualization, Methodology, Software, Validation. **Y. Elloumi:** Funding acquisition, Project administration, Supervision, Writing – original draft. **M. Akil:** Investigation, Writing – review & editing. **M.H. Bedoui:** Visualization, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial or personal relationships that could be viewed as influencing the work reported in this paper.

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